Hardware User Manual

AnyDIO Series

SIO-DO32P

AJINEXTEK AXT

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Revision History

Manual	PCB	Comments
Rev. 1.0 issue 1.0	Rev. 1.0	2001. 12. 18

1. Outline

1.1.Introduction

AnyDIO module board is loaded on the AnyBus Series board. It has an input module, output module and input-output module which can control digital signal's input-output, and it is composed of various product group providing up to 32 points of IO contacts per each module. Among the models of AnyDIO module boards, the SIO-DO32P module board provides 16 points input/ 16 points output (power TR output mode) of digital IO contacts.

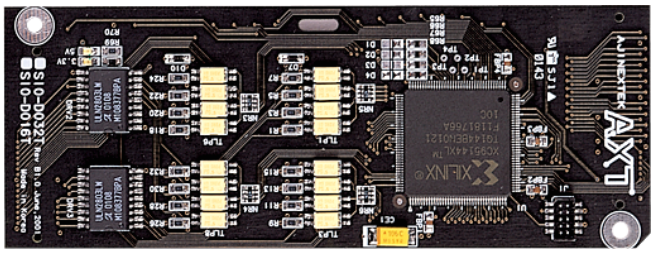
1.2.Application

AnyDIO module board can interface the external digital I/O signals, and can be used on the factory automation, machine tool control, laboratory work, educational purpose, etc.

- Digital Monitoring
- Signal Conversion
- Optical Insulation between Computer and the Peripheral
- Automatic Test Equipment
- Equipment Monitoring
- Equipment Interface
- Process Control
- Industrial Measurement
- Industrial Digital Output Control

2. Features

2.1.Module Specifications and Scheme



Picture 1. Real Picture of SIO-DO32P Module Board

Table 1. Features of SIO-DO32P Module Board

Items	SIO-DO32P
Output	32 points
	32 points output can be written or read in bit and byte units.
Range of Output Voltage	12~24Vdc
Output Current	50 ㎃ (up to 68Pin CHAMP connector) Option : 50 ㎃ per each pin when using T36-PR terminal block. 500 ㎃ per each pin when using APC-EO36 terminal block.
Output Resistance	Open collector type
Output Circuit	Sink (Negative logic)
	Every output port is electrically isolated with the external signal so the external signal can be driven. Isolation makes use of a photo coupler.

Table 2. Specifications of SIO-DO32P Module Board

# of Output Channel	16 points
Output Range	External power license mode (12V ~ 24V)
Output Method	Open collector type
Output Mode	Voltage
Turn ON/OFF Time	20㎲ and over
Active Indicator	2 LEDs for the license confirmation (5VDC, 3.3VDC)
Others	Isolated with photo coupler
	Characteristics of isolation: 2500VRMS

3. Installation

3.1.Hardware Installation

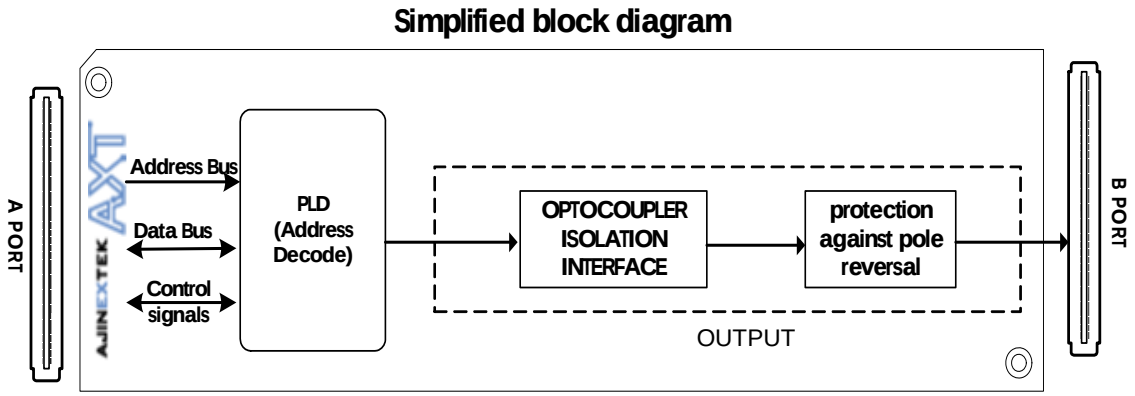
The loading direction of an AnyPack module (SIO-DO32P) in Add-On type varies according to the basic mode baseboard a user selected. Load a module properly with reference to 'Hardware Description' of the baseboard user manual and then install according to the order of 'Hardware Installation'.

3.2.Software Installation

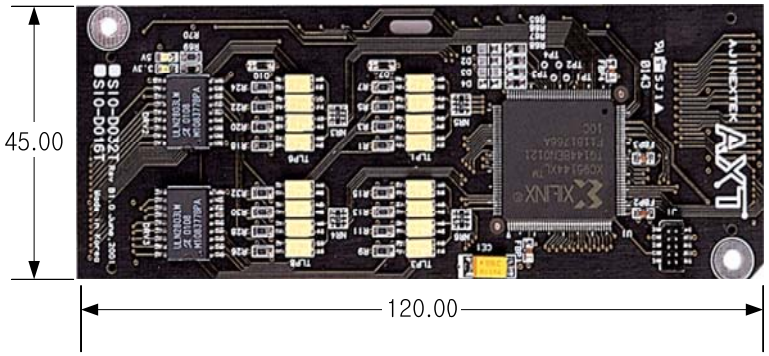
Install EzConfig referring to the EzConfig 1.0 user manual. If an installation target folder is not assigned yet, it is presumed that installation of EzDioAgent program is normally completed when the EzDioAgent.exe executive file exists in the C:\Program Files\EzConfig. If the installation target folder is assigned separately, check whether the EzDioAgent.exe executive file exists in the assigned folder or not.

4. Configuration

4.1.Hardware Configuration and Description



Picture 2. Block Diagram of SIO-DO32P Module Board



Picture 3. Size of SIO-DO32P Board

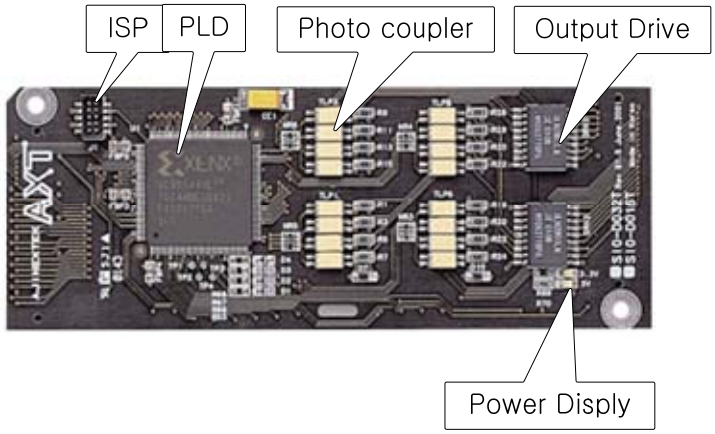
Table 3. Environment in Use of SIO-DO32P Module Board

Board Size	120 x 45 mm
Working Voltage	3.3V, 5V, 24V (12V)

Depending on how you combine AnyDIO modules on the AnyBus board, the number of control IO contacts can be extended or modified by a simple operation in the control system using various IO devices.

If BPFR(PCI Full Size) is selected as a basic mode baseboard, it can compose up to 128 points with 4 AnyDIO

modules having 32 contacts. And it can be composed of 128 points input board, 64 points input/64 points output board and 128 points output board according to the selection of AnyDIO series. If SUB1 is used as a CPU slot on the BPFR board, it can get 96 DIO contacts by loading up to 32 points of AnyDIO on the 3 SUB slots. This configuration method can be changed into 96 points input (three SIO-DI32), 96 points output (three SIO-DO32P), 48 points input/48 points output (three SIO-DB32P), etc. depending on which AnyDIO module is used.

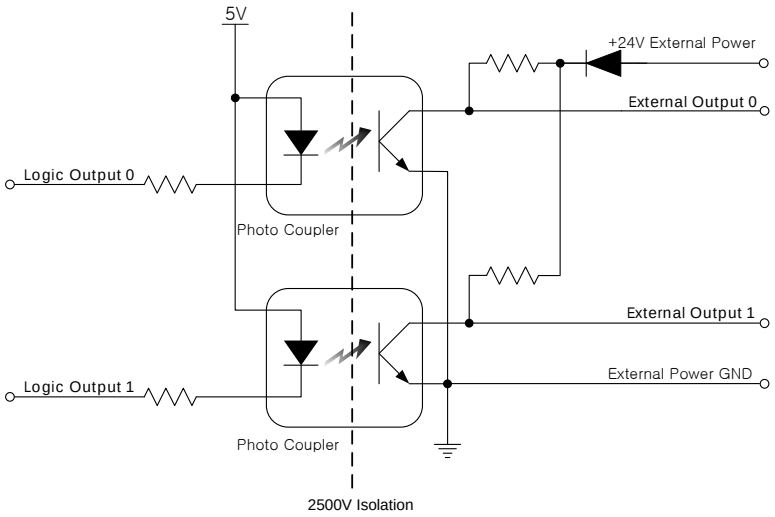


Picture 4. Configuration of SIO-DO32P

Picture 4 shows a control division (PLD), SUB module connector, photo coupler, output drive, indication LED, etc. of the SIO-DO32P.

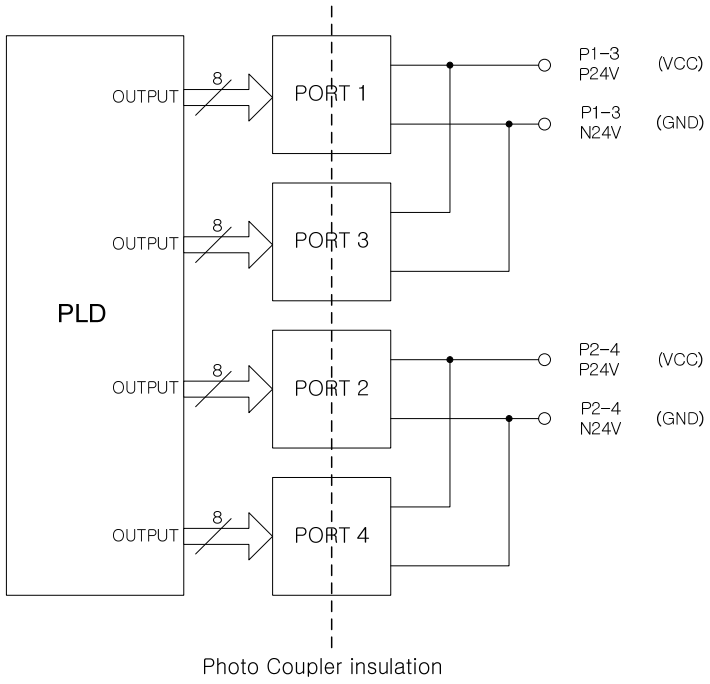
There are two wide functions in the control division. First, it has a function to write 32 points output. Second, it controls a data flow between the basic mode baseboard and the local module board. The first function implies that output control provides a latch in order to write the external data of the basic mode baseboard so the external data can be transmitted smoothly. Second function is that it separates registers by internal functions through the decoder so it makes data transmission between the basic mode baseboard and the local module easy.

64 pin PMC(PCI Mezzanine Card) SUB module connectors of the A port and the B port are summarized in short as follows. The A port supplies 5V and 3.3V power to module from the AnyBus baseboard. The signal line of the A port is composed of data 8 bit, address 9 bit, read and write signals, selection signal of module, one interrupt request signal (IRQ) and so on. The B port receives the external power supply and 32 points output digital IO contacts are connected with the 68 CHAMP connector.



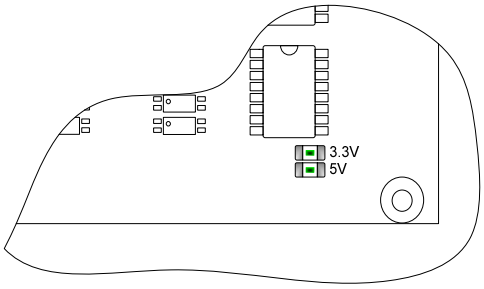
Picture 5. Output Signal's Photo Coupler

All the 32 points digital outputs are isolated with the external signals by the photo coupler.



Picture 6. Power Processing inside of the Module

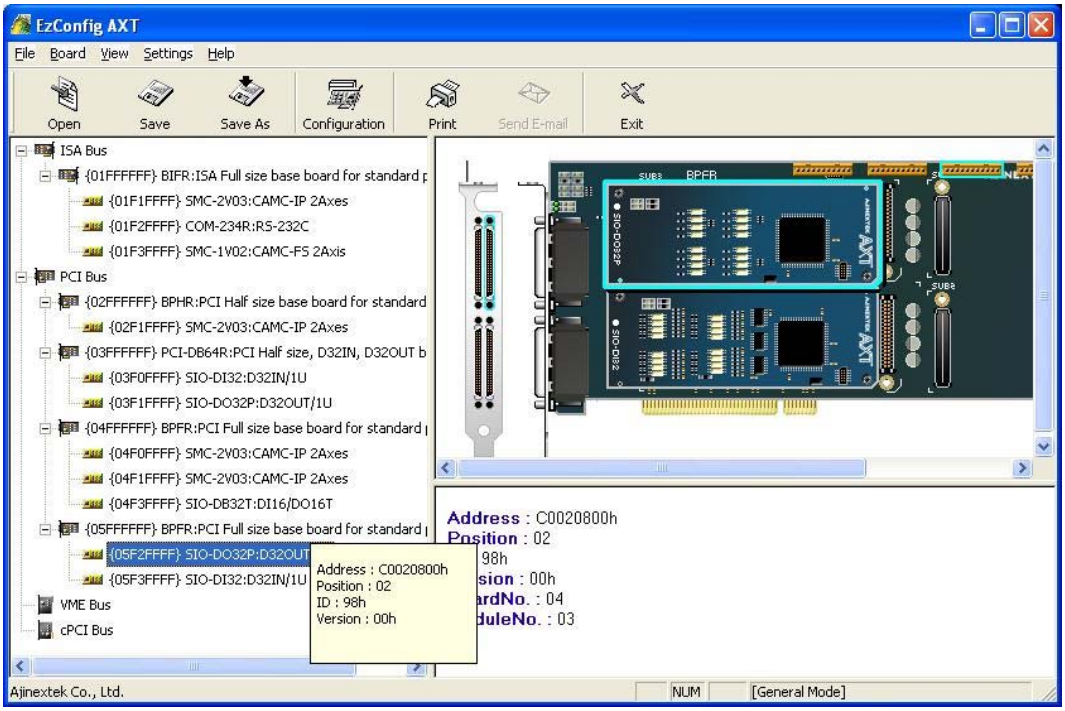
Picture 6 shows the power process inside of module. Each port is composed of 8 bits. Port1 and Port3 are connected with the same power supply line, and so are port2 and port4. Therefore it is easy to separate from equipment or sensor that makes a lot of noise by using each different power supplier.



Picture 7. Power Indication LED

Two LEDs on the SIO-DO32P module board are attached in order to indicate getting a supply of 3.3V and 5V power from the AnyBus baseboard. The LED is turned on if power is supplied from the AnyBus base and turned off if power is not supplied.

4.2.Module Address Map



Picture 8. SIO-DO32P Information View Screen Using EzConfig Software

Table 4. Range of Address Allocated to Each SUB

Region	BIFR, BV6R, BPFR, BC6R (FULL SIZE / 6U)	BIHR, BV3R, BPHR, BC3R (HALF SIZE / 3U)	Remarks
	0000h – 1FFFh	0000h – 1FFFh	
0000h – 03FFh	Address space allocated to SUB1	Address space allocated to SUB1	1Kbyte
0400h – 07FFh	Address space allocated to SUB2	Address space allocated to SUB2	1Kbyte
0800h – 0BFFh	Address space allocated to SUB3	-	1Kbyte
0C00h – 0FFFh	Address space allocated to SUB4	-	1Kbyte

If you select BIFR (ISA Full Size) board as a basic mode baseboard and start viewing information of SIO-DO32P module loaded on the SUB2 by using EzConfig software, you can get information that an address of module2 is the initial address of SIO-DO32P.

Address = Base Address + 0400h(SUB 2 Offset Address)

If you press baseboard (B) tab on the information view dialog window, you can get information about the baseboard easily.

4.2.1. Address Map

Table 5. Address Map of SIO-DO32P Module Board

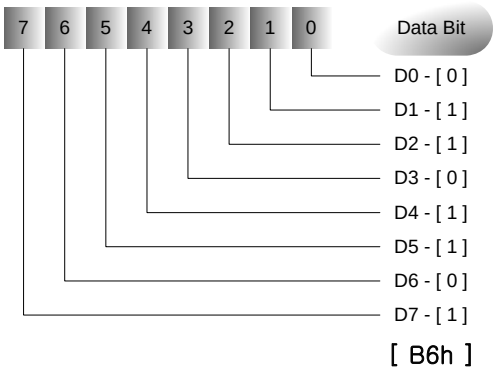
Offset	Function	Data	Length	Direction	Initial Value
00h	Preamble		8 bit	R	B6h
02h	ID		8 bit	R	98h (152)
04h	Version		8 bit	R	00h
06h	Software Reset	Bit0 : 1(ON) Software reset when write	1 bit	R/W	00h
..0Fh	Reserved				
10h	Output port 1	Bit0..7 : 0(OFF)/1(ON)	8 bit	R/W	00h
12h	Output port 2	Bit0..7 : 0(OFF)/1(ON)	8 bit	R/W	00h
14h	Output port 3	Bit0..7 : 0(OFF)/1(ON)	8 bit	R/W	00h
16h	Output port 4	Bit0..7 : 0(OFF)/1(ON)	8 bit	R/W	00h
20h	Reserved				
22h	Reserved				
24h	Reserved				
26h	Reserved				
30h	Reserved				
32h	Reserved				
34h	Reserved				
36h	Reserved				
40h	Reserved				
42h	Reserved				
44h	Reserved				
46h	Reserved				
50h	Reserved				
52h	Reserved				
54h	Reserved				
56h	Reserved				
60h	Reserved				
62h	Reserved				

The SIO-DO32P address has a space of 1kbyte, and the actual space used is from 00h to 16h.

4.2.2. Functional Description

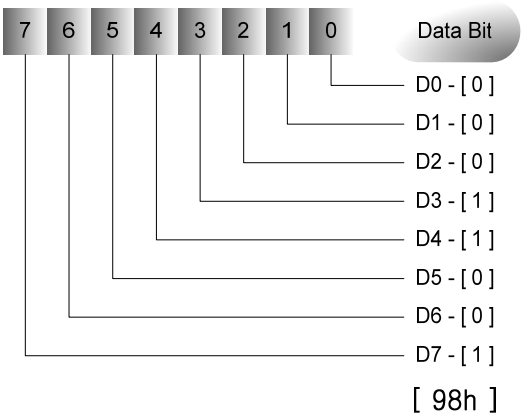
A. Preamble Register : Read Only(00h)

Preamble register arranged the combination of ‘0’ and ‘1’ and the value of ‘10110110’(0xB6) is read. This value is always read in this register address, so it is used on the automatic cognition of a module board together with the ID register.



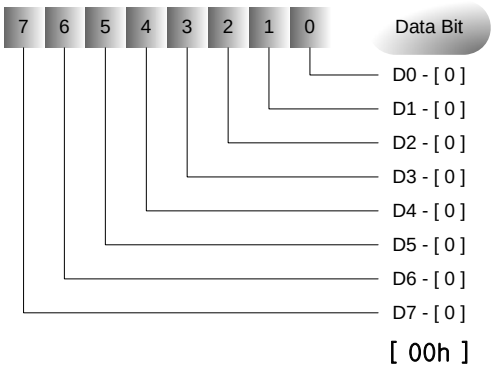
B. ID Register : Read Only(02h)

Each module has its own ID(identifier), so this information can distinguish which module board was loaded on the current baseboard. The ID of a SIO-DB32P module board is ‘10011000’(0x98h)이다.



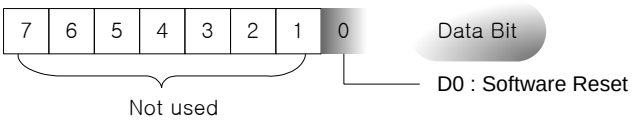
C. Version Register : Read Only(04h)

The current information on the firmware and the hardware mounted on the current module board can be captured. If you read this register address, you can also read the version of SIO-DO32P.



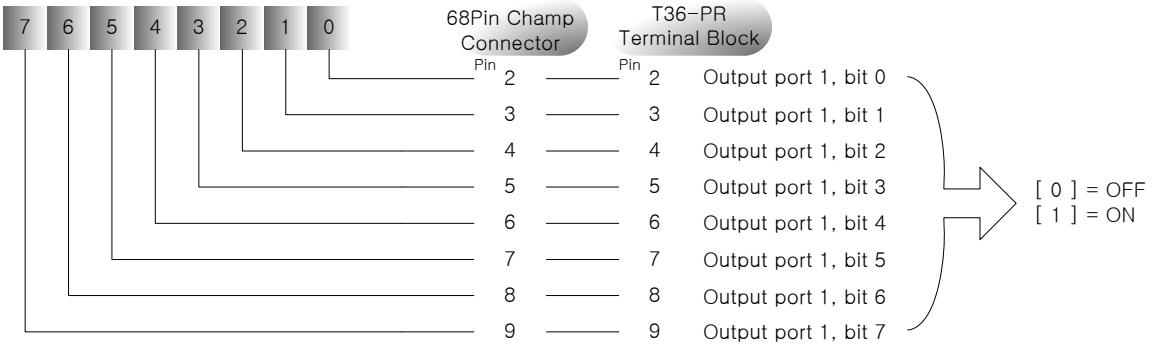
D. Software Reset : Read Only(06h)

Using of a baseboard's reset switch or booting of a PC can initialize the state of the module board. And if you write '1' on the lowest rank bit D0 of the 06h address on the program, every register phase can be also fixed as the initial value listed in the table 5. That is, it functions as a reset switch of the baseboard.



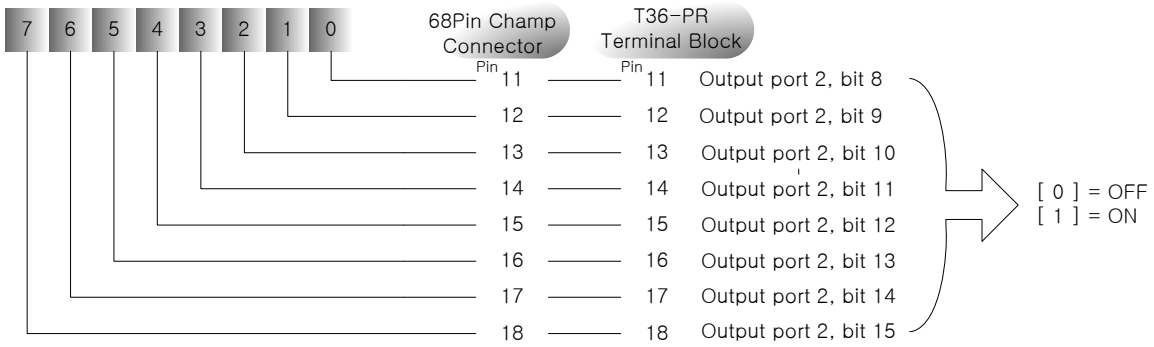
E. Output Port1 Register : Read and Write(10h)

The user can generate a special data value from the outside only through this register. Since the user can read (read back) data wrote on the Output Port1 Register again, the value wrote can be confirmed.



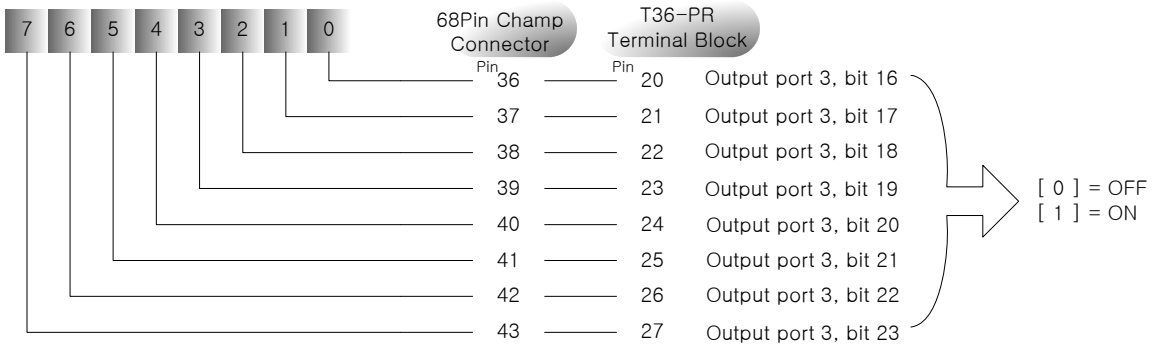
F. Output Port2 Register : Read and Write(12h)

The user can generate a special data value from the outside only through this register. Since the user can read (read back) data wrote on the Output Port2 Register again, the value wrote can be confirmed.



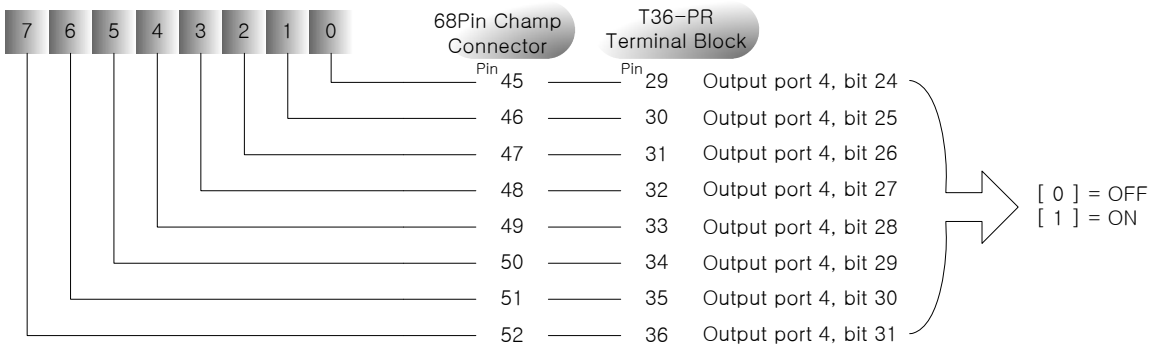
G. Output Port3 Register : Read and Write(14h)

The user can generate a special data value from the outside only through this register. Since the user can read (read back) data wrote on the Output Port3 Register again, the value wrote can be confirmed.



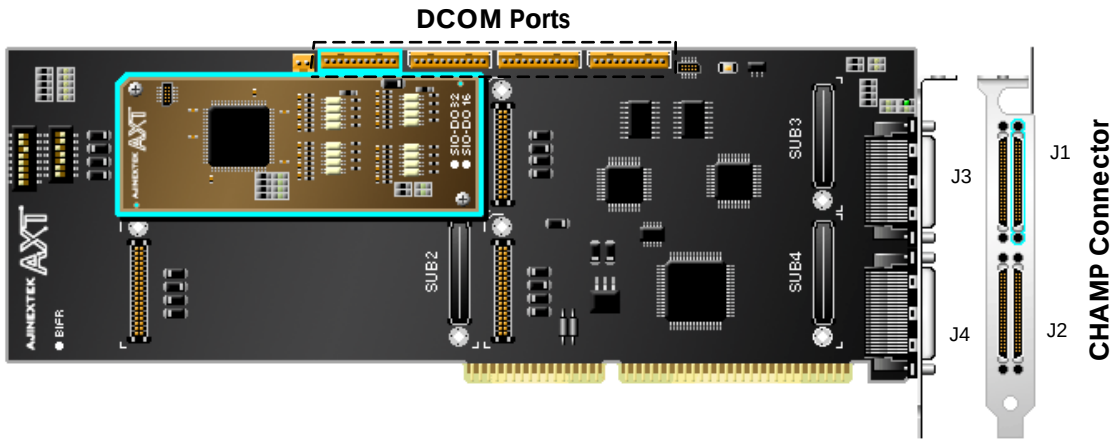
H. Output Port4 Register : Read and Write(16h)

The user can generate a special data value from the outside only through this register. Since the user can read (read back) data wrote on the Output Port2 Register again, the value wrote can be confirmed.



4.3.Pin Array and Description of SIO-DO32P Module Board Champ Connector

SIO-DO32P is a digital output board having 32 points output, and it is electrically isolated with the external power connected through the connector. When you connect the external device and the terminal block, be sure to refer to the following table.



Picture 9. In Case of Loading SIO-DO32P on the BIFR's SUB1

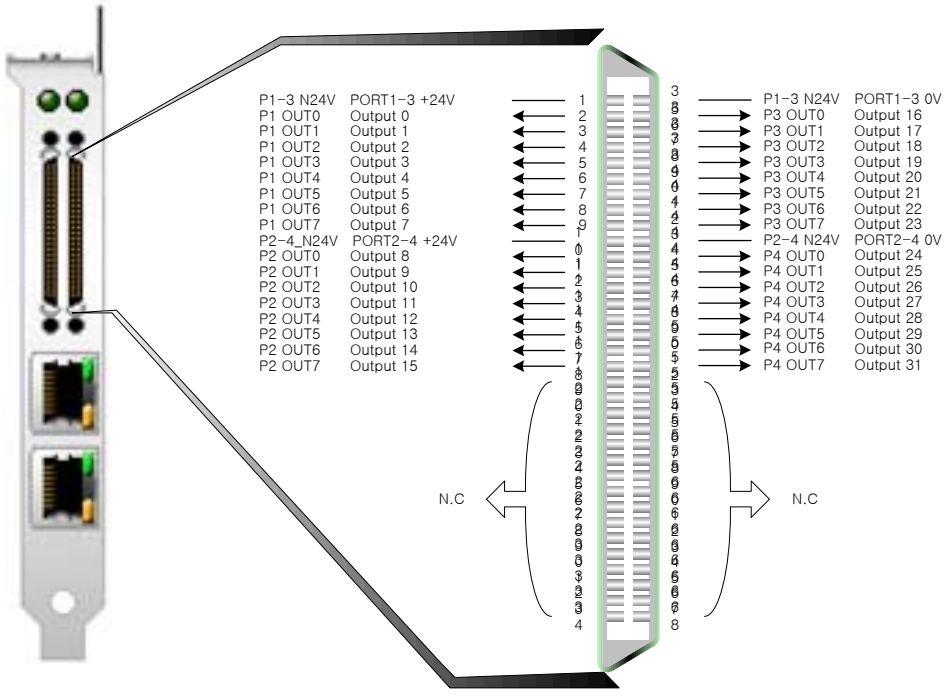
Picture 9 shows the case loaded with SIO-DO32P on the SUB1/CPU module socket of the BIFR board. The external IO interface division connected to the outside through the special or general purpose cable is composed of a 68pin CHAMP connector and a DCOM Port, a distributed master module connector, and each 68pin connector and DCOM Port are connected with the SUB module. DCOM Port is used when AnyField Series products are loaded, so this manual will not make the subject here.

Table 6. 68Pin CHAMP Connector Pin Array When SIO-DO32P Module is Loaded

J1	Pin Name	Dir.	Description	J1	Pin Name	Dir.	Description
1	P1-3_P24V		+24V Power of PORT1	35	P1-3_N24V		0V Power of PORT-3
2	P1_OUT0	O	Outport 1, bit0	36	P3_OUT0	O	Outport 3, bit 16
3	P1_OUT1	O	Outport 1, bit1	37	P3_OUT1	O	Outport3, bit 17
4	P1_OUT2	O	Outport 1, bit2	38	P3_OUT2	O	Outport 3, bit 18
5	P1_OUT3	O	Outport 1, bit3	39	P3_OUT3	O	Outport 3, bit 19
6	P1_OUT4	O	Outport 1, bit4	40	P3_OUT4	O	Outport 3, bit 20
7	P1_OUT5	O	Outport 1, bit5	41	P3_OUT5	O	Outport 3, bit 21
8	P1_OUT6	O	Outport 1, bit6	42	P3_OUT6	O	Outport 3, bit 22
9	P1_OUT7	O	Outport 1, bit7	43	P3_OUT7	O	Outport 3, bit 23
10	P2-4_P24V		+24V Power of PORT2-4	44	P2-4_N24V		0V Power of PORT2-4
11	P2_OUT0	O	Outport 2, bit8	45	P4_OUT0	O	Outport 4, bit 24
12	P2_OUT1	O	Outport 2, bit9	46	P4_OUT0	O	Outport 4, bit 25
13	P2_OUT2	O	Outport 2, bit10	47	P4_OUT0	O	Outport 4, bit 26
14	P2_OUT3	O	Outport 2, bit11	48	P4_OUT0	O	Outport 4, bit 27
15	P2_OUT4	O	Outport 2, bit12	49	P4_OUT0	O	Outport 4, bit 28
16	P2_OUT5	O	Outport 2, bit13	50	P4_OUT0	O	Outport 4, bit 29
17	P2_OUT6	O	Outport 2, bit14	51	P4_OUT0	O	Outport 4, bit 30
18	P2_OUT7	O	Outport 2, bit15	52	P4_OUT0	O	Outport 4, bit 31
19	N.C			53	N.C		
20	N.C			54	N.C		

J1	Pin Name	Dir.	Description	J1	Pin Name	Dir.	Description
21	N.C			55	N.C		
22	N.C			56	N.C		
23	N.C			57	N.C		
24	N.C			58	N.C		
25	N.C			59	N.C		
26	N.C			60	N.C		
27	N.C			61	N.C		
28	N.C			62	N.C		
29	N.C			63	N.C		
30	N.C			64	N.C		
31	N.C			65	N.C		
32	N.C			66	N.C		
33	N.C			67	N.C		
34	N.C			68	N.C		

J1, J2, J3, J4 are connected with the SUB1, SUB2, SUB3, SUB4, respectively, and they are connected to the terminal block or the external device through a cable. Therefore, 68pin CHAMP connector J1's signals are arranged as table 6 when SIO-DO32P module is loaded on the SUB1 socket. Among the 68 pins, a total of 36 pins including 32 points output and 4 power sources are used only.



Picture 10. CHAMP Connector

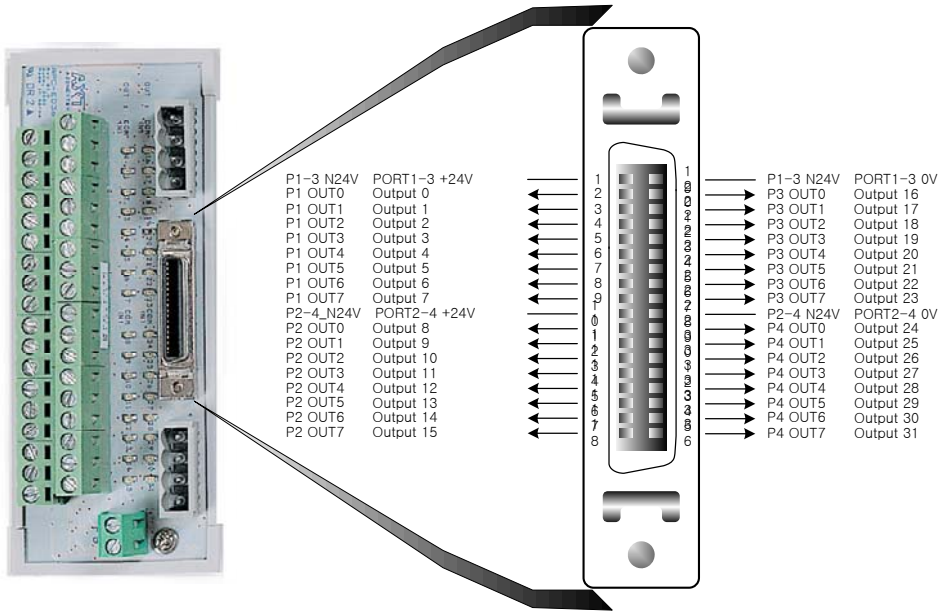
Picture 10 shows the 68 CHAMP connector's pin connected to the SIO-DO32P module.

4.4.Pin Connection and Description of SIO-DO32P Module Board Terminal Block

Table 7. Pin Connection and Description of T36-PR, APC-EB36 Terminal Block

T36-PR APC-EB36	Pin Name	Dir.	Description	T36-PR APC-EB36	Pin Name	Dir.	Description
1	P1-3_P24V		+24V Power of PORT-3	19	P1-3_N24V		0V Power of PORT-3
2	P1_OUT0	O	Outport 1, bit0	20	P3_OUT0	O	Outport 3, bit 16
3	P1_OUT1	O	Outport 1, bit1	21	P3_OUT1	O	Outport 3, bit 17
4	P1_OUT2	O	Outport 1, bit2	22	P3_OUT2	O	Outport 3, bit 18
5	P1_OUT3	O	Outport 1, bit3	23	P3_OUT3	O	Outport 3, bit 19
6	P1_OUT4	O	Outport 1, bit4	24	P3_OUT4	O	Outport 3, bit 20
7	P1_OUT5	O	Outport 1, bit5	25	P3_OUT5	O	Outport 3, bit 21
8	P1_OUT6	O	Outport 1, bit6	26	P3_OUT6	O	Outport 3, bit 22
9	P1_OUT7	O	Outport 1, bit7	27	P3_OUT7	O	Outport 3, bit 23
10	P2-4_P24V		+24V Power of PORT3-4	28	P2-4_N24V		0V Power of PORT2-4
11	P2_OUT0	O	Outport 2, bit8	29	P4_OUT0	O	Outport 4, bit 24
12	P2_OUT1	O	Outport 2, bit9	30	P4_OUT1	O	Outport 4, bit 25
13	P2_OUT2	O	Outport 2, bit10	31	P4_OUT2	O	Outport 4, bit 26
14	P2_OUT3	O	Outport 2, bit11	32	P4_OUT3	O	Outport 4, bit 27
15	P2_OUT4	O	Outport 2, bit12	33	P4_OUT4	O	Outport 4, bit 28
16	P2_OUT5	O	Outport 2, bit13	34	P4_OUT5	O	Outport 4, bit 29
17	P2_OUT6	O	Outport 2, bit14	35	P4_OUT6	O	Outport 4, bit 30
18	P2_OUT7	O	Outport 2, bit15	36	P4_OUT7	O	Outport 4, bit 31

This table shows the pin connection and description from a viewpoint of terminal block when the SIO-DO32P module is loaded.



Picture 11. Definition of Terminal Block 36 Pin Connector Pin

5. Accessories (Optional Items)

5.1.Module Accessories and Configuration

The optional cable and terminal block used to connect the AnyDIO module and the external device are as follows.

Table 8. Terminal Block and Cable When AnyDIO Module is Loaded

Module Name	Terminal Block	Cable
SIO-DI32	APC-EI36 or T36-PR	C68/36TS
SIO-DO32P	APC-EO36 or T36-PR	C68/36TS
SIO-DB32P	APC-EB36 or T36-PR	C68/36TS
SIO-DI16 SIO-DO16T SIO-DB16T	Not in use	No private cable

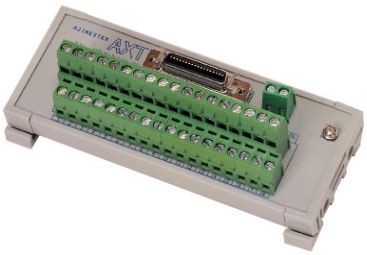
A. Cable



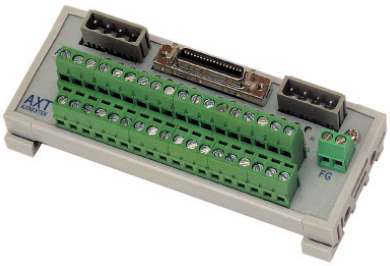
Picture 12. C68/36TS

C68/36TS, an asymmetric cable, has a Pitch (pin interval) of 0.8mm and 1.27mm. This cable connects the external connector (J1,J2,J3,J4) of the AnyBus baseboard and the terminal block one-to-one. The twist pair consists of 68 pins, such as 1 and 35, 2 and 36, …., 34 and 68. Max. 50mA of current can be flowed per each pin. Each cable is 1m, 2m, 3m long in length, and it can be supplied up to 5m long on the user's demand.

B. Terminal Block



T36-PR One-to-One Terminal Block



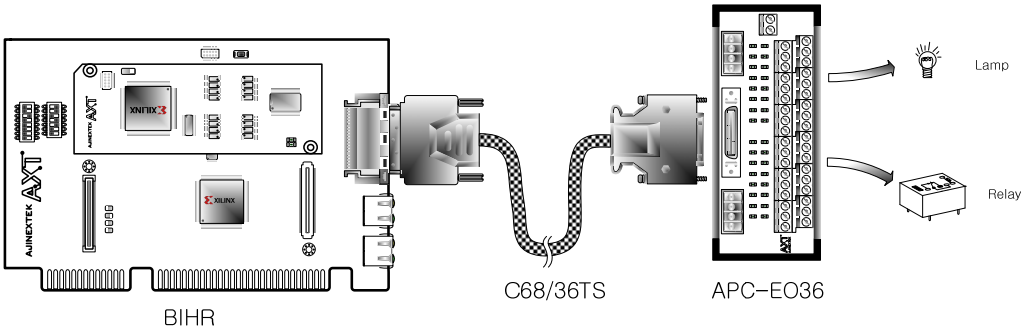
APC-EB36 Digital I/O Terminal Block

Picture13. Types of Terminal Block – One-to-One Terminal Block (left) and High Quality I/O Terminal Block (right)

In the AnyDIO module, a terminal block (APC-EO36) with the current amplified circuit inside or a general terminal block (T36-PR) connected one-to-one can be used on the output edge.

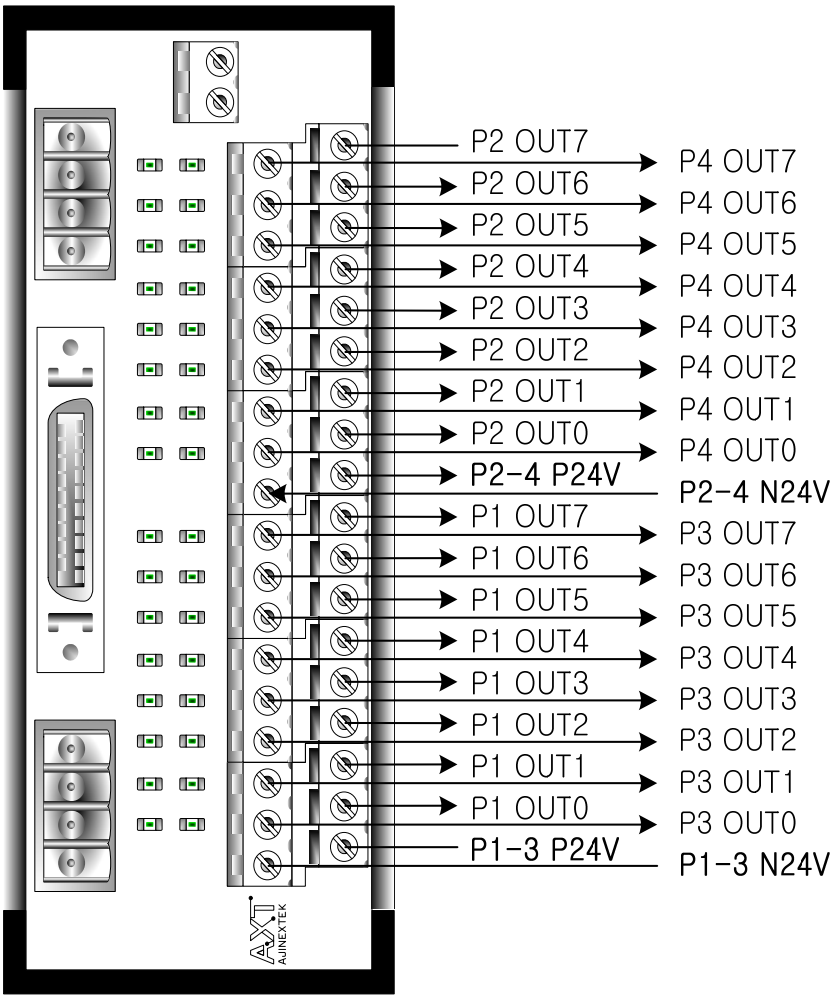
T36-PR and APC-EO36 terminal blocks are designed to be able to load on the 35mm wide slide type DIN-Rail, so it is easy to take on/off in case of installation and wiring work.

5.2.Device Connection



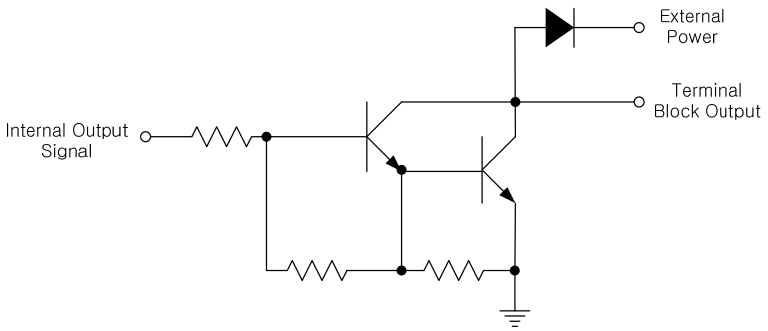
Picture 14. Connection between SIO-DO32P Module Board and the Peripheral (Terminal Board)

Picture 14 shows the connection between C68/36TS cable and APC-EO36 terminal block that are used to connect the external devices when the SIO-DO32P module board is loaded on the real BIHR board. And a diagram shows the connectable sensors to the outside of a terminal block. In addition to the parts illustrated above, it is able to interface some other parts the user wants.



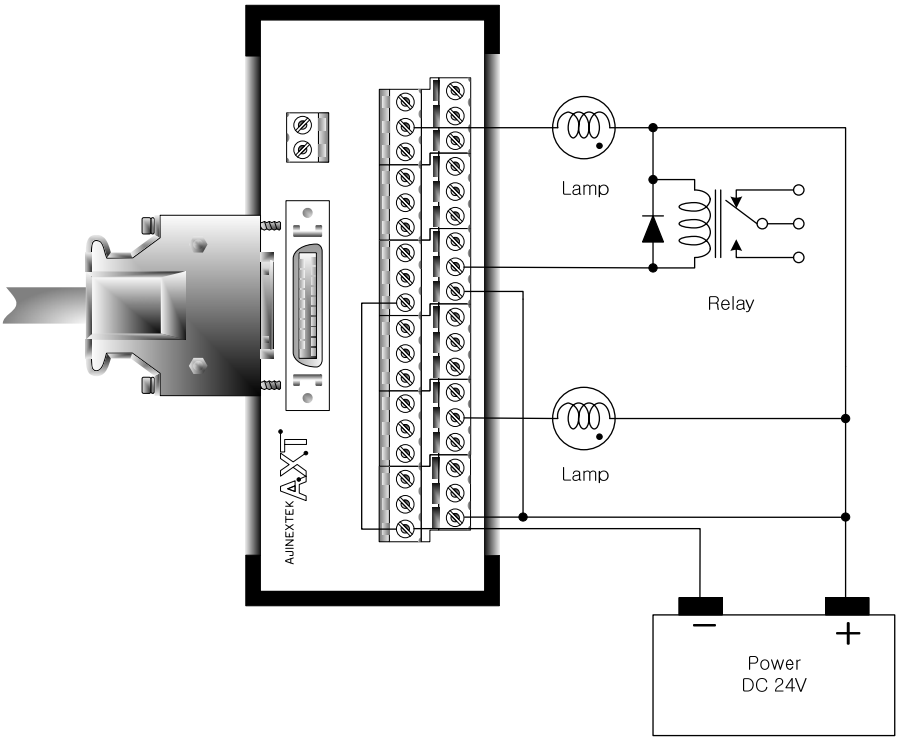
Picture 15. Description of Terminal Block Pin

Picture 15 shows the pin name and location of the terminal block.



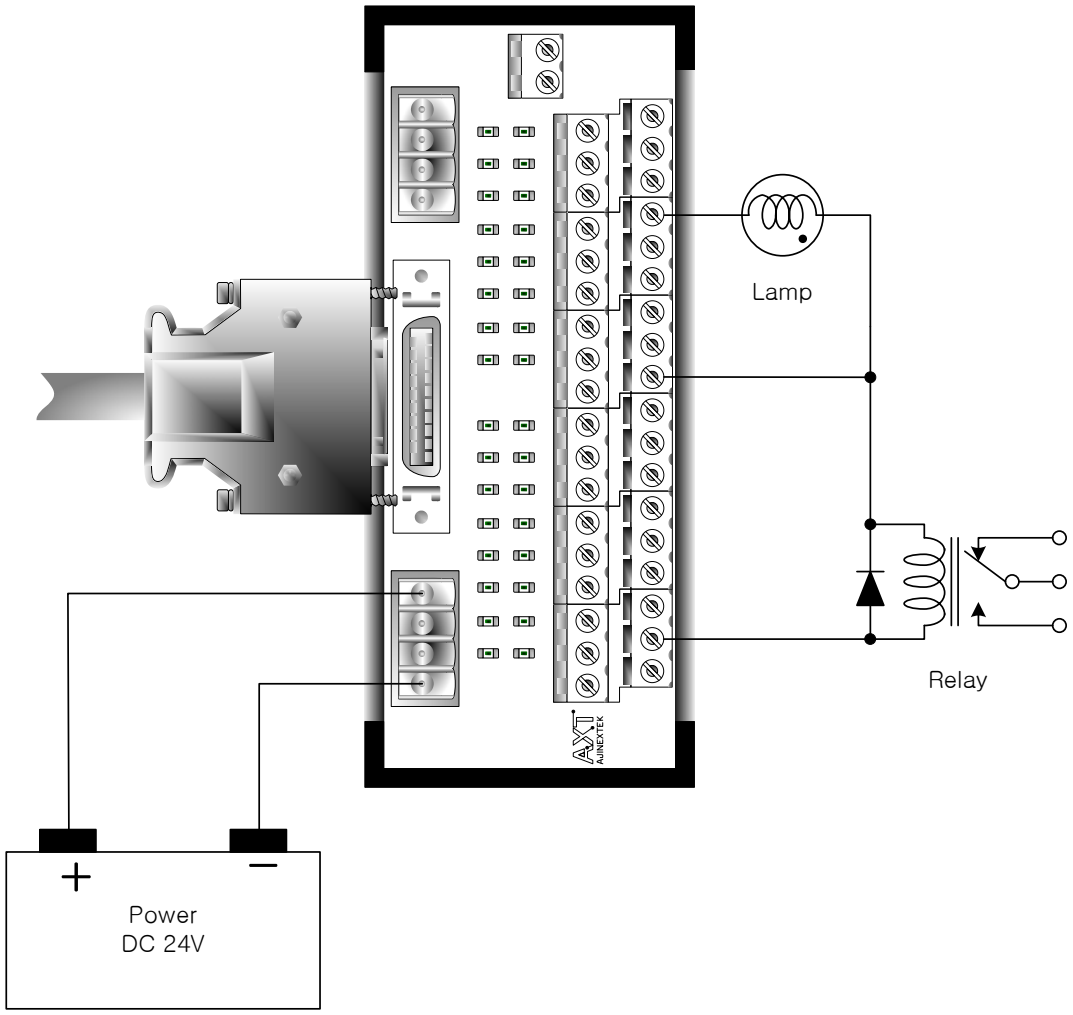
Picture 16. Output Circuit Diagram inside of Terminal Block

Picture 16 shows output circuit diagram of the terminal block.



Picture 17. Basic Configuration Using T36-PR Terminal Block

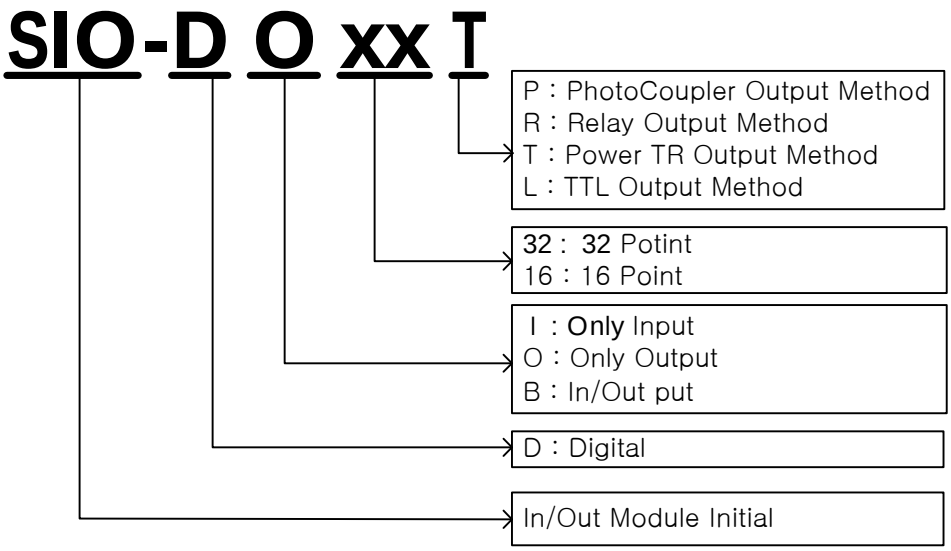
Picture 17 shows a wiring diagram between T36-PR terminal block and components. The lamp is connected to the PORT1 – OUT4, PORT4-OUT6 and the relay is connected to the PORT2 – OUT0. T36-PR can send output current up to 50mA. Output is operated by sink (negative logic).



Picture 18. Basic Configuration Using APC-EO36 Terminal Block

Picture 18 shows a wiring diagram between APC-EO36 terminal block and components. The lamp is connected to the PORT2 – OUT4, and the relay is connected to the PORT1 – IN0. APC-EO36 terminal block has two power connectors outside, so power connection between terminal blocks can be used helpfully when terminal blocks are used a lot. The output current can pass up to 500mA per each pin, output is operated by sink (negative logic).

6. Ordering Information (Family)



Picture 19. The Way of Reading AnyDIO Series Model Name

Table 9. AnyDIO Module's Product Group

Product Group	Description
SIO-DI32	(Centralized) 32 Points Digital Input Module
SIO-DI16	(Distributed) 16 Points Digital Input Module
SIO-DO32P	(Centralized) 32 Points Digital Output Module
SIO-DO16T	(Distributed) 16 Points Digital Output Module
SIO-DB32P	(Centralized) 16 Points Digital Output/ 16 Points Digital Input, Power TR (2803) Output
SIO-DB16T	(Distributed) 8 Points Digital Output/ 8 Points Digital Input, Power TR (2803) Output

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